0. Cover

- **Project title:** Protly - Sequence-to-insight protein analysis
- **Team name & ID:** T-8
- **Institute / Course:** GLA University
- **Version:** v1.0
- **Date:** 5 Feb 2026

Revision history

Version	Date	Author	Change
v0.1	5 Feb 2026	Jyotsana Bhardwaj	Initial draft

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1. Overview

- **Problem statement:** There is an explosion of known protein sequences in public databases, but only a small fraction of these proteins have been experimentally characterized for their structure and function. Wet-lab analysis is slow, costly, and requires specialized biological expertise, which creates a bottleneck for researchers, students, and engineers who want quick, reliable insights from protein sequences. Existing bioinformatics tools are often fragmented, not AI-native, and demand domain knowledge, making them less accessible to non-experts.
- **Goal:** Build an AI-driven web platform that accepts a protein by name or raw amino acid sequence and generates an integrated "computational intelligence report" including function, metadata, lab-relevant physicochemical properties, and structure visualization. The system will leverage protein language models (ESM-2 / ProtBERT), transformer-based representations, and downstream neural "heads" to predict solubility, stability, and subcellular localization directly from sequence.
- **Non-goals:** In the current version, Protly will not perform de novo structure prediction from scratch (it will rely on precomputed AlphaFold DB structures where available). It will not include full-fledged experimental design workflows, in vitro validation, or wet-lab automation. Advanced generative

protein design and large-scale mutational scanning will be treated as future scope rather than core deliverables for v1.

- **Value proposition:** Protly treats amino acid sequences as a language and uses transformer-based PLMs to learn the "grammar" of protein folding and function without requiring a Ph.D. in biology. It unifies hybrid search (name and sequence), UniProt knowledge, AI predictions, Biopython-derived lab metrics, and AlphaFold 3D visualization into a single, user-friendly dashboard. This enables fast, scalable, and accessible protein analysis for students, researchers, and engineers in computational proteomics.

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2. Scope and Control

2.1 In-scope

• Hybrid search bar with toggle for:

- Name-based search using UniProt REST API to resolve protein names (e.g., "Human Insulin") to primary accession IDs.

- Sequence-based search using semantic similarity over PLM embeddings to find functionally related proteins.

• Integration with UniProt API to fetch: biological function summary, source organism, and gene name for the selected protein ("Protein Biography").

• Embedding layer using ESM-2 / ProtBERT to convert sequences into high-dimensional vectors capturing structural and functional information.

• Task-specific head models on top of embeddings for: solubility, stability, and subcellular localization prediction (e.g., trained/evaluated on datasets like DeepLoc).

• Physicochemical metric computation via Biopython: instability index, GRAVY score, and isoelectric point (pI).

• Integration with AlphaFold Protein Structure Database API for retrieving precomputed 3D structures and confidence metrics where available.

• Web dashboard (e.g., Streamlit) presenting search, analysis results, and basic structure viewer embedding.

2.2 Out-of-scope

• Custom training of large-scale PLMs from scratch; Protly will rely on publicly available pretrained models like ESM-2.

• Molecular dynamics simulations, docking workflows, or drug discovery pipelines.

• Regulatory validation for clinical diagnostics or medical decision support.

• Full-featured LIMS (laboratory information management system) and large batch job orchestration in v1.

2.3 Assumptions

• Stable internet connectivity is available to query UniProt, AlphaFold DB, and any remote PLM endpoints if used.

• Typical sequences lie within the supported length of the chosen PLM (e.g., ESM-2 variants generally support sequences up to around 1,022 residues without special handling).

• Users are familiar with basic molecular biology concepts (e.g., amino acids, proteins, pH) but may not have deep expertise in bioinformatics.

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3. Stakeholders and RACI

Activity	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Requirements	Rudraksh Mahajan	Rudraksh Mahajan	Mentor	Team
Architecture & AI pipeline	Siddhartha Khandelwal	Rudraksh Mahajan	Mentor	Team
Implementation (backend, APIs, PLM integration)	Jyotsana Bhardwaj	Rudraksh Mahajan	Mentor	Team
Implementation	Rudraksh Mahajan	Rudraksh Mahajan	Mentor	Team

(dashboard UI & visualization)				
Testing & Documentation	All team members	Rudraksh Mahajan	Mentor	Team

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4. Team and Roles

Member	Role	Responsibilities	Key skills	Availability	Contact
Rudraksh Mahajan	Team Leader	Finalize scope, align requirements, coordinate milestones, demo preparation	Python, ML, APIs, project coordination	8 hrs/wk	rudraksh.mahajan_cs.h23@gla.ac.in
Siddhartha Khandelwal	AI/Backend Engineer	Integrate PLMs (ESM-2/ProtBERT), build embedding and head models, implement UniProt & AlphaFold API calls, backend services	Python, PyTorch, REST APIs, bioinformatics basics	10 hrs/wk	siddhartha.khandelwal_cs.h23@gla.ac.in
Jyotsana Bhardwaj	Frontend & Visualization Engineer	Design and develop a comprehensive dashboard, implement hybrid search UI, integrate plots and 3D viewer,	Python, Streamlit, basic JS/HTML, UI design	10 hrs/wk	Jyotsana.bhardwaj_cs.h23@gla.ac.in

		documentation support			

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5. Week-wise Plan and Assignments

Week	Dates	Milestones	Rudraksh (Lead)	Siddhartha (Backend)	Jyotsana (Frontend)	Deliverables	Status
1	1-7 Feb	Requirements freeze	Finalize scope	API contracts v0	Wireframes	Draft SRS	Planned
2	8-20 Feb	Architecture & DB	Arch review	Schema & migrations	UI kit	ERD, API spec v1	Planned
3	21-28 Feb	Analyzing	Risks, unblock	Auth + Users API	Auth screens	Auth module	Planned
4	1-10 Mar	Frontend scaffolding	Sync	Listings API	Routing, forms	UI shells	Planned
5	11-22 Mar	Feature set A	Cut scope	Create/Edit listing E2E	Listing pages	Feature A demo	Planned
6	23-31 Mar	Feature set B	KPIs	Search API	Search UI + Profile	Feature B demo	Planned
7	1-7 April	Hardening	Risk burn-down	Bug fixes	Bug fixes	Test report	Planned
8	8-15 April	Release & deck	Final sign-off	Release notes	Polish UI	v1.0, slides	Planned

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6. Users and UX

6.1 Personas

- **Research Student:** Wants quick understanding of protein function, localization, and lab properties.

- **Computational Biologist:** Wants to screen sequences for stability and solubility before wet-lab experiments.

6.2 Top user journeys

- Journey 1 (Name-based): Home → Select "Name" → Enter "Human Insulin" → Display biography, predictions, metrics, 3D structure.
- Journey 2 (Sequence-based): Home → Select "Sequence" → Paste sequence → Generate intelligence report.

6.3 User stories

- As a student, I want to enter a protein name and get an easy-to-read biography plus AI-predicted properties.
- As a researcher, I want to paste a custom sequence and estimate stability, solubility, and localization.

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7. Key Features

- Hybrid Search Engine: Central search bar with toggle for Name vs Sequence search (Must Have).
- Protein Biography: Auto-generated panel showing biological function, organism, gene name (Must Have)
- AI Property Heads: Neural models predicting stability, solubility, localization (Must Have)
- Physicochemical Suite: Biopython calculations and hydropathy plots (Should Have)
- Structural Visualization: AlphaFold 3D viewer with confidence coloring (Should Have)

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8. Objectives and Success Metrics

- **O1 Classification Accuracy:** Achieve competitive accuracy for subcellular localization on held-out test set. KPI: Overall accuracy and F1 score for localization classes.

- **O2 Performance and Latency:** p Ensure dashboard returns complete report within 5 seconds for typical sequences. KPI: p95 response time ≤ 5 seconds
- **O3 Usability:** Seamless toggle between search modes with minimal friction. KPI: Users complete search and view report in ≤ 3–4 clicks.
- **O4 Reliability:** Robust handling of invalid inputs and missing structures. KPI: No critical crashes during demo; clear error messages for 95% of edge cases.

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9. Delivery & Operations

- **Release Plan** v1.0 ready for internal demo by end of planned semester timeline.
- **Deployment:** Initially on a local lab server or single cloud VM with Python environment.
- **Monitoring:** Lightweight logging of requests, errors, and latency for debugging and performance tuning.

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10. Research and Evaluation

- Literature review on protein language models (ESM-2, ProtBERT) and their use in structure/function prediction.
- Study of AlphaFold DB APIs and integration patterns in web tools.
- Evaluation: Quantitative metrics on head models (F1 scores) and qualitative usability feedback from faculty/peers

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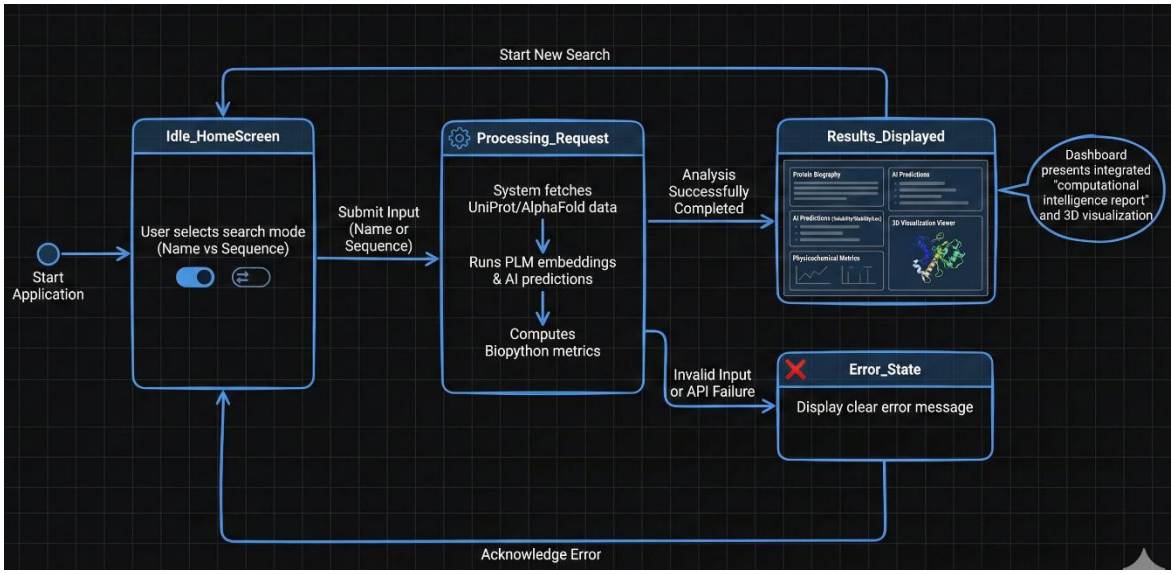
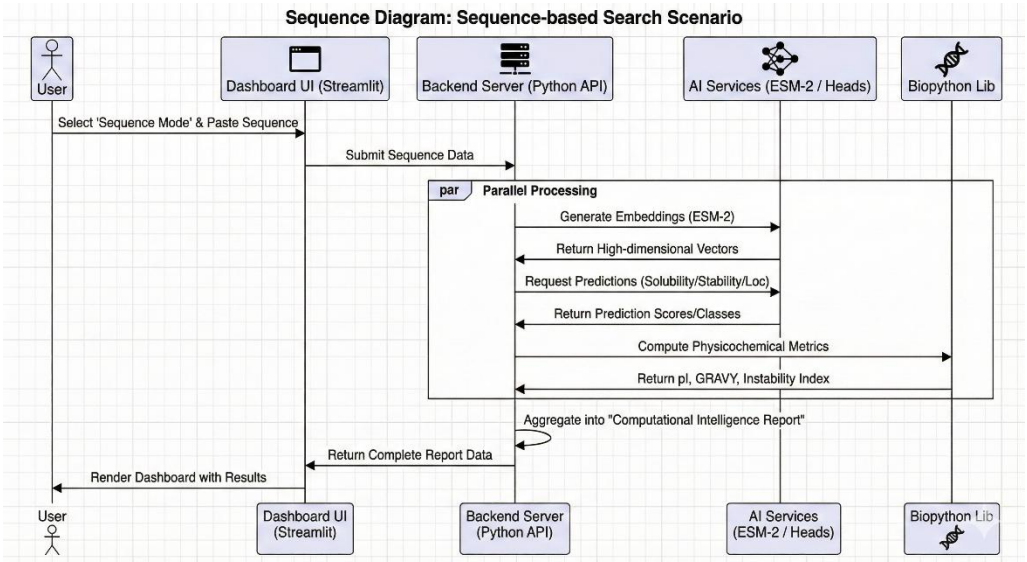
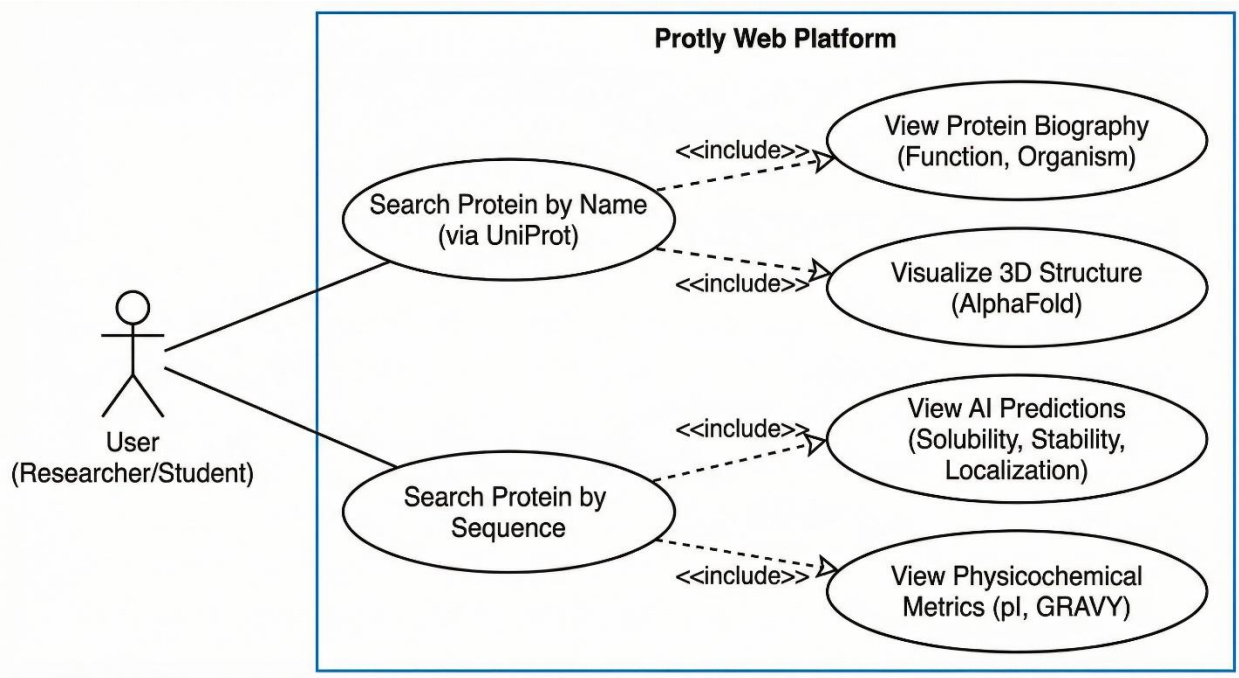
11. Technical Workflow Diagram



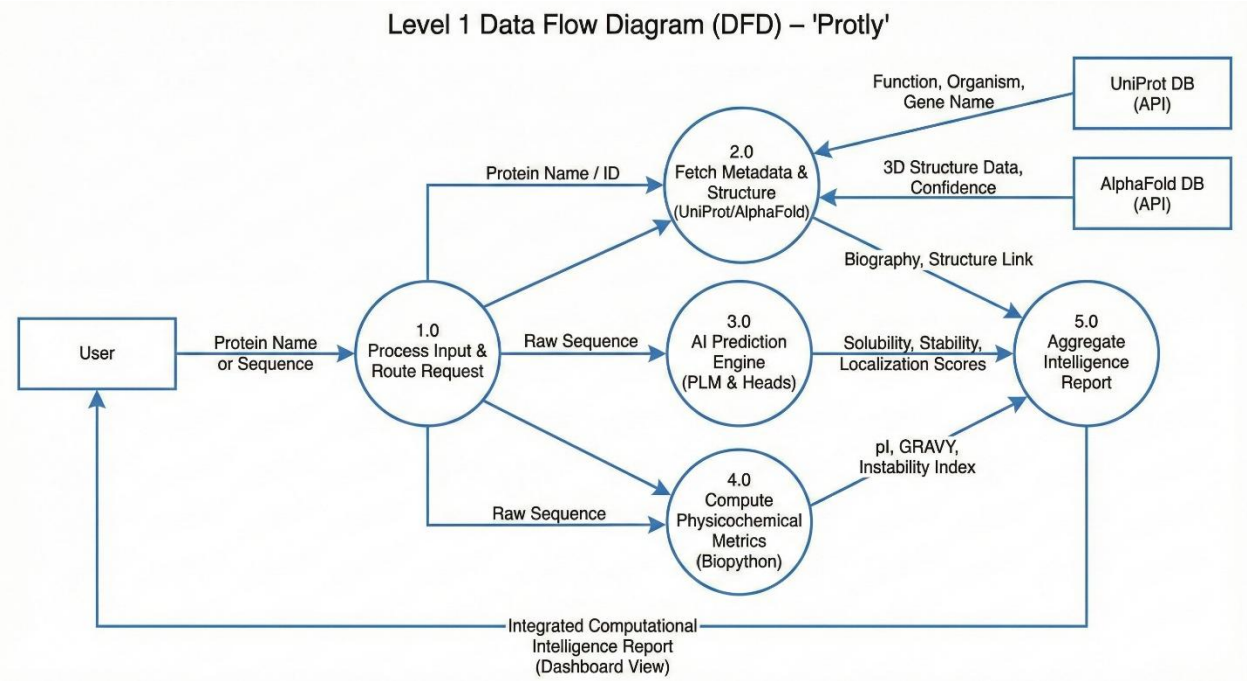
i. State Transition Diagram



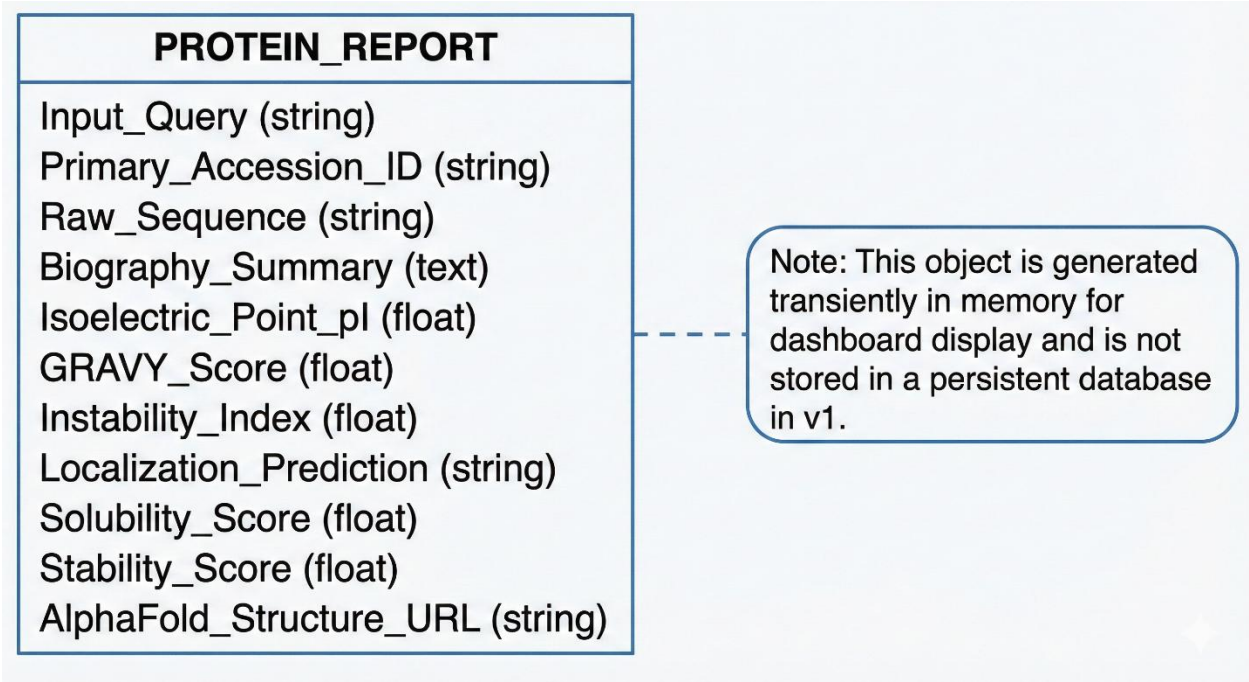
ii. Sequence Diagram



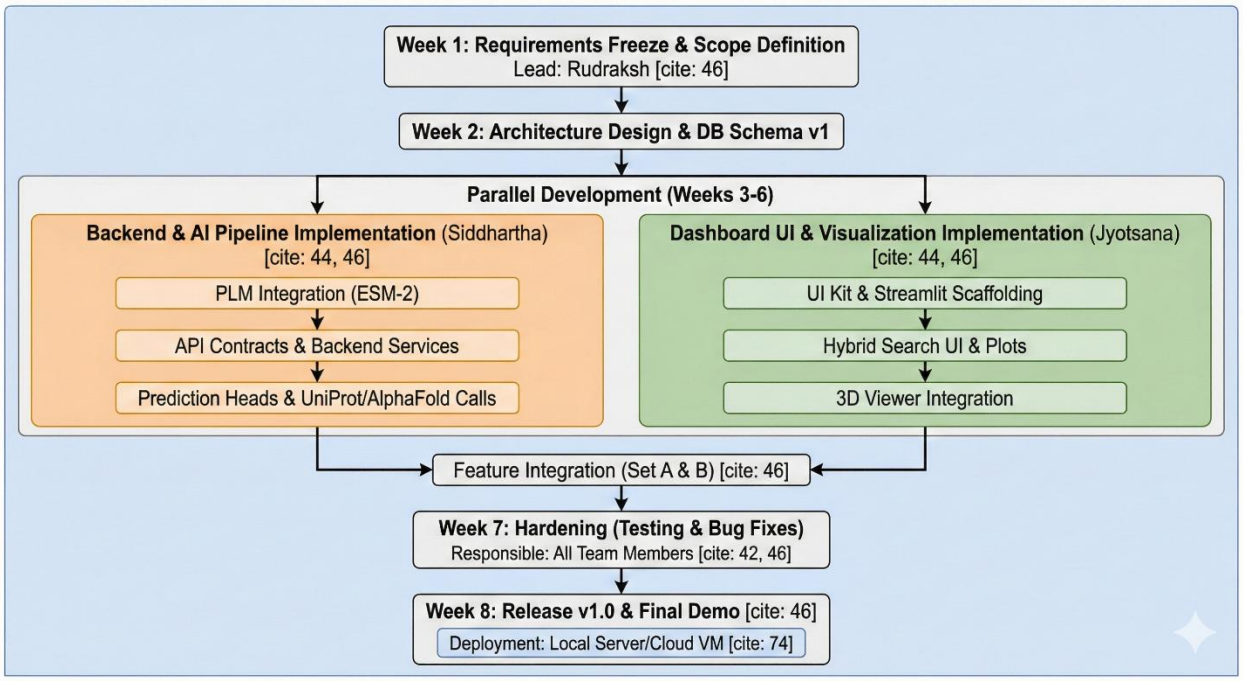
iii. Use Case Diagram



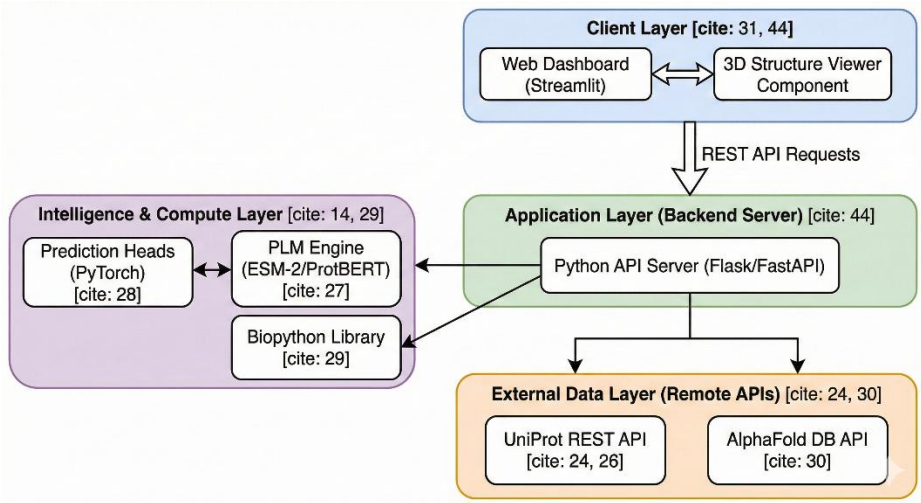
iv. Data Flow Diagram



v. ER Diagram



vi. Technical Workflow Diagram



vii. Work Architecture Diagram

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12. Appendices

- **Glossary:** AI, AlphaFold DB, Biopython, ESM-2, GRAVY Score, Instability Index, KPI, p95, PLM, ProtBERT, RACI.
- **References:**
  - https://youtu.be/GHoE4VkDehY?si=KiECvjNuNhTdm-nv, https://www.uniprot.org/
  - https://en.wikipedia.org/wiki/Protein_structure
  - https://en.wikipedia.org/wiki/Protein_structure_prediction
  - https://alphafold.ebi.ac.uk/
  - https://cen.acs.org/biological-chemistry/proteomics/Microsoft-AI-predicts-protein-conformations/103/web/2025/07
  - https://www.microsoft.com/en-us/research/blog/from-static-prediction-to-dynamic-characterization-ai2bmd-advances-protein-dynamics-with-ab-initio-accuracy/
  - https://www.npmjs.com/
  - https://en.wikipedia.org/wiki/API